

# Cibersegurança / Cybersecurity

(METI 1º Ano/S2)

## Network Security Technologies

Henrique Santos ([hsantos@dsi.uminho.pt](mailto:hsantos@dsi.uminho.pt))

Dpt. Sistemas de Informação

Ext. 3302

# Index

## Preliminary considerations

- Main threats:
  - Remote access and/or modification of programs and data (includes credential theft)
  - Remote program execution (*malware*)
  - Interception and/or modification of data in transit
  - Data insertion in communications
  - Lost of integrity in DNS
  - Communication interruption (total or selective)
- New attacks explore several vulnerabilities and are more multifaceted!

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## Firewalls

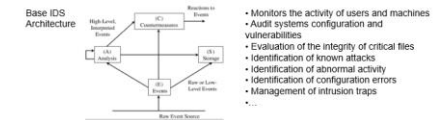
- Officially invented in 1990, but based on the concept of reference monitor from the years 70
- Basically it is a device that filters network traffic protecting an internal network from an untrusted "outside world"
  - Set of basic features:
    - Impossible to circumvent – analyzes all traffic (performance is critical)
    - Focuses its function at the level 2 of network (IP header); but may include features of level 3 – flow control through the SN and the TCP ports → **Connection Tracking**
    - Resistant to attacks – minimal functionality
    - Low complexity – facilitates the analysis and administration

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## Intrusion Detection Systems (IDS)

- **Firewalls** and Access Control prevent malicious activity
- What about bad "legitimate" utilization? It must be **detected** ... ⇒ IDS
  - Typically, a computer system looking for "suspected" activity

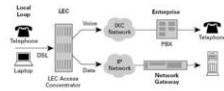


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## Notes about VPN

- Secure remote access
  - Before, Data over Voice (DoV) – distance-dependent cost
  - Today Data over Data (DoD) – single tariff, just depending on the bandwidth and traffic. Advantages:
    - Economic factors
      - Number of users
      - Sharing of resources (WAN)
    - Flexibility and scalability
    - Bandwidth
    - Any "last mile" technology



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## Notes about IPSec

- Promotes: **confidentiality, integrity and authenticity**
- Standard defined by the IETF for secure communication ("real-time")
- Assumes that the parties share a session key (distributed manually, or with IKE-below)
- IPSec works with both IPv4 and IPv6
- SA (**Security Association**) denotes an **unidirectional connection cryptographically secure** (address, sequence numbers, encryption algorithms, ensures or not integrity, etc.). Each SA has a key and a unique identification, SPI – **Security Parameter Index**
- The IPSec header identifies the respective SA
- Any system that implements IPSec has to have a database of SAs (**SAD** – Security Association Database)
- Each host must be configured with all necessary parameters, which are stored in a **SPD** – **Security Policy Database**

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## Notes about TLS/SSL

- SSL (Secure Sockets Layer) – Protocol created by Netscape
- Positioned between application and transport levels (TCP), and ensures:
  - Server authentication
  - Optional client authentication (more versatile in TLS, using TLS-PSK, a shared key mechanism)
  - Encrypted channel with shared key (negotiated algorithm – 3DES, RC4, etc); **symmetric key generation algorithm more resistant in the TLS**
- Protocol summary:
  - Client requests an SSL session
  - Server responds with its digital certificate (public key)
  - The "**Cipher Suite**" is negotiated
  - Client sends a seed for the session key, encrypted with the server's public key, and they both calculate the session key
  - Optionally the messages are authenticated (hash function – HMAC – in TLS; pseudo random function in SSL)

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# Preliminary considerations

- Main threats:
  - Remote access and/or modification of programs and data (includes credential theft)
  - Remote program execution (*malware*)
  - Interception and/or modification of data in transit
  - Data insertion in communications
  - Lost of integrity in DNS
  - Communication interruption (total or selective)
- New attacks explore several vulnerabilities and are more multifaceted!

# Preliminary considerations

- First line of defence: Management and infrastructure planning
- Second line of defence: filtering (*firewalls*)
- Third line of defence: monitoring and intrusion detection
- Fourth line of defence: encryption...

# Technological Stack for Information Security

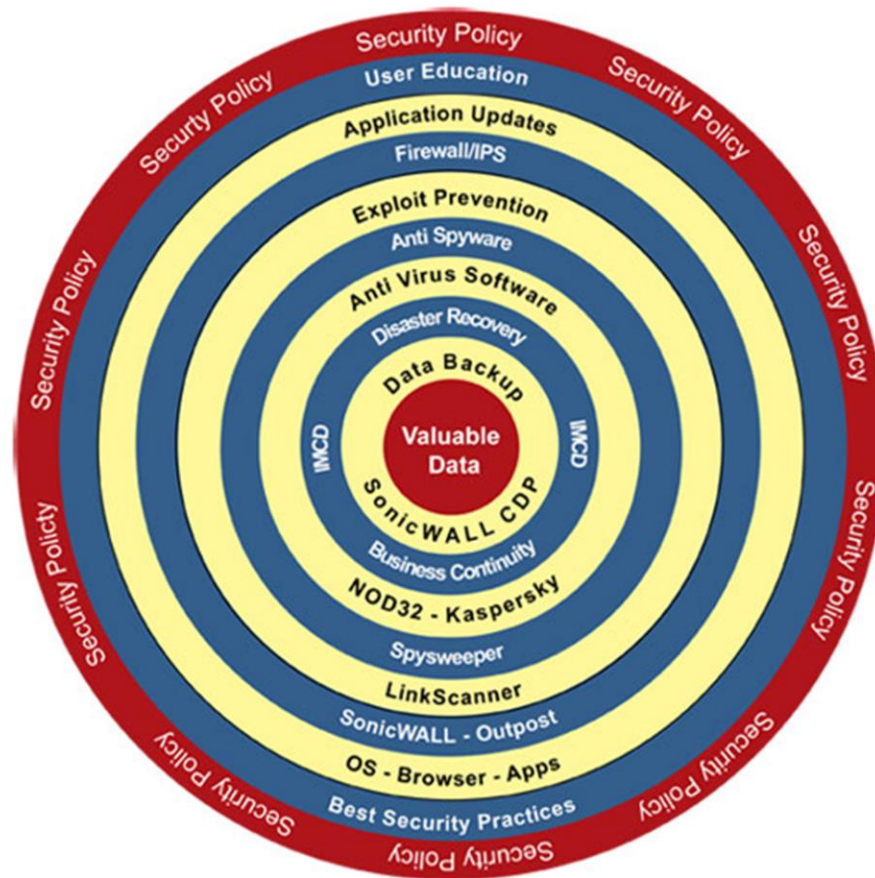


Still, network traffic covers a substantial number of attacks!

<https://mitre-attack.github.io/attack-navigator/>

# Management and infrastructure planning

- ***Defense in Depth***: It is a common strategy used to secure subnets



Software Security Solutions Layered Security Model <sup>TM</sup>

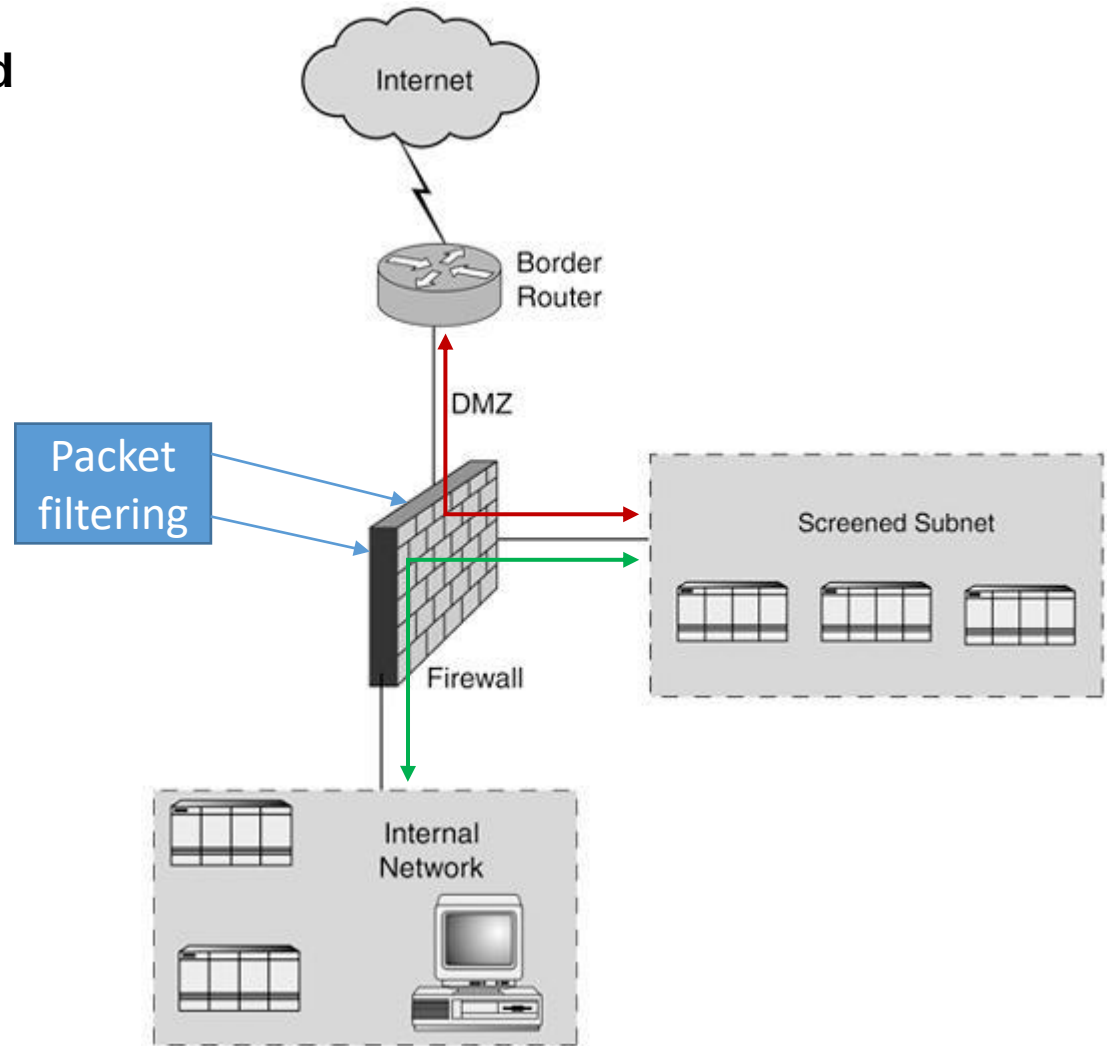
# Management and infrastructure planning

- **Baseline Approach**  
(ISO/IEC 13335-4)
- Several proposals...

|                                                | Stand-alone Workstation | Workstation (Client without Shared Resources) Connected to a Network | Server or Workstation with Shared Resources Connected to a Network |
|------------------------------------------------|-------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>I&amp;A</b>                                 |                         |                                                                      |                                                                    |
| I&A Based on Something the User Knows          | X                       | X                                                                    | X                                                                  |
| I&A Based on Something the User Possesses      | X                       | X                                                                    | X                                                                  |
| I&A Based on Something the User Is             | (X)                     | (X)                                                                  | (X)                                                                |
| <b>Logical Access Control and Audit</b>        |                         |                                                                      |                                                                    |
| Access Control Policy                          |                         |                                                                      | X                                                                  |
| User Access to Computers                       | X                       | X                                                                    | X                                                                  |
| User Access to Data, Services and Applications | X                       | X                                                                    | X                                                                  |
| Reviewing and Updating Access Rights           |                         |                                                                      | X                                                                  |
| Audit Logs                                     | X                       | X                                                                    | X                                                                  |
| <b>Malicious code</b>                          |                         |                                                                      |                                                                    |
| Scanners                                       | X                       | X                                                                    | X                                                                  |
| Integrity Checkers                             | X                       | X                                                                    | X                                                                  |
| Removable Media Circulation Control            | X                       | X                                                                    | X                                                                  |
| Procedural Safeguards                          | X                       | X                                                                    | X                                                                  |
| <b>Network Management</b>                      |                         |                                                                      |                                                                    |
| Operational Procedures                         |                         |                                                                      | X                                                                  |
| System Planning                                |                         |                                                                      | X                                                                  |
| Network Configuration                          |                         |                                                                      | X                                                                  |
| Network Segregation                            |                         |                                                                      | X                                                                  |
| Network Monitoring                             |                         |                                                                      | X                                                                  |
| Intrusion Detection                            |                         |                                                                      | X                                                                  |
| <b>Cryptography</b>                            |                         |                                                                      |                                                                    |
| Data Confidentiality Protection                | (X)                     | (X)                                                                  | (X)                                                                |
| Data Integrity Protection                      | (X)                     | (X)                                                                  | (X)                                                                |
| Non-Repudiation                                |                         | (X)                                                                  | (X)                                                                |

# Management and infrastructure planning

- **De-Militarized Zones and Screened Subnets**

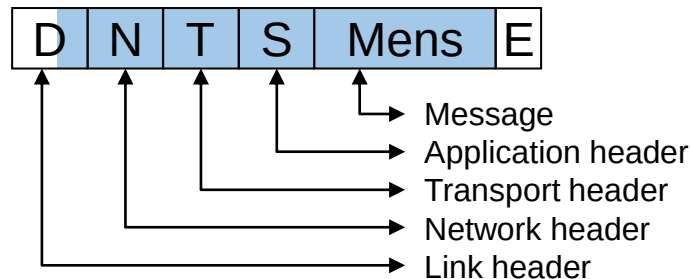




# Management and infrastructure planning

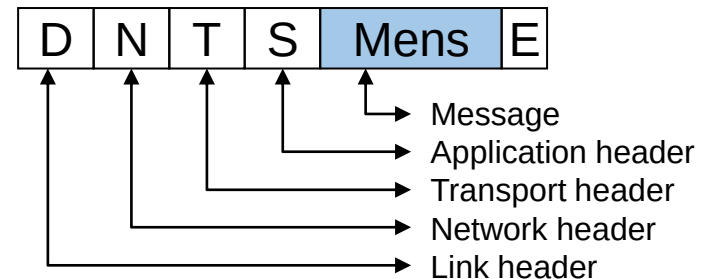
- Network architecture (preventive control)
  - Promote the segmentation (facilitates the monitoring and management)
  - Promoting redundancy
  - Avoid the singular points of failures
  - **Promote encrypted communications (risk-based)**

- *Link Encryption*



- When the transmission medium is the focus of threats

- *End-to-End Encryption*



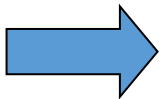
- When computers are the focus of threats

# Firewalls

- Officially invented in 1990, but based on the concept of reference monitor from the years 70
- Basically it is a device that filters network traffic protecting an internal network from an untrusted “outside world”
  - Set of basic features:
    - Impossible to circumvent – analyzes all traffic (performance is critical)
    - Focuses its function at the level 2 of network (IP header); but may include features of level 3 – flow control through the SN and the TCP ports → **Connection Tracking**)
    - Resistant to attacks – minimal functionality
    - Low complexity – facilitates the analysis and administration

# Firewalls

- Typically do not reach the application level (proxies are better suited)
- Several implementation strategies  $\Rightarrow$  Appropriate security policy
- Attention to the default behaviour:



- What is not expressly prohibited is **accepted**; or
- What is not expressly accepted it is **forbidden**

# Firewalls

- Packet processing engine implemented in Linux

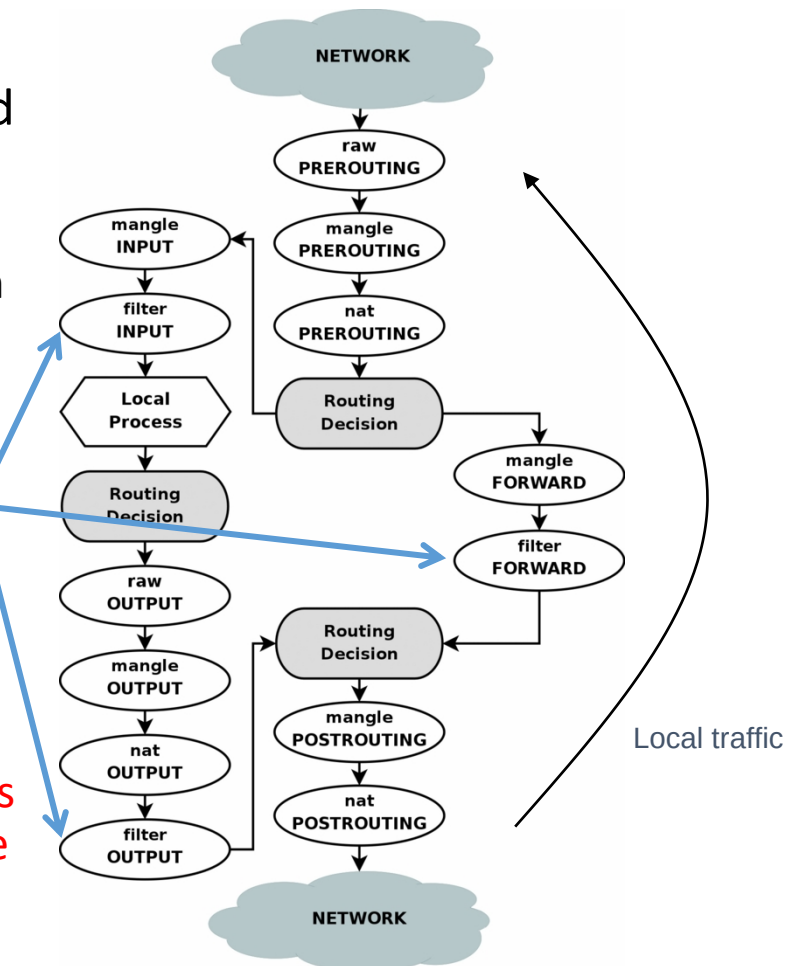
- Each of the 5 **chains** are associated with two or more **tables** ("**queues**"), each with sequences of rules that define the global behaviour of the "*machine*"

- The **firewall** must execute in the **table filter**, in any of the chains (by definition)

- In the *prerouting*, it is not advisable to do filtering, since it may involve modification of headers

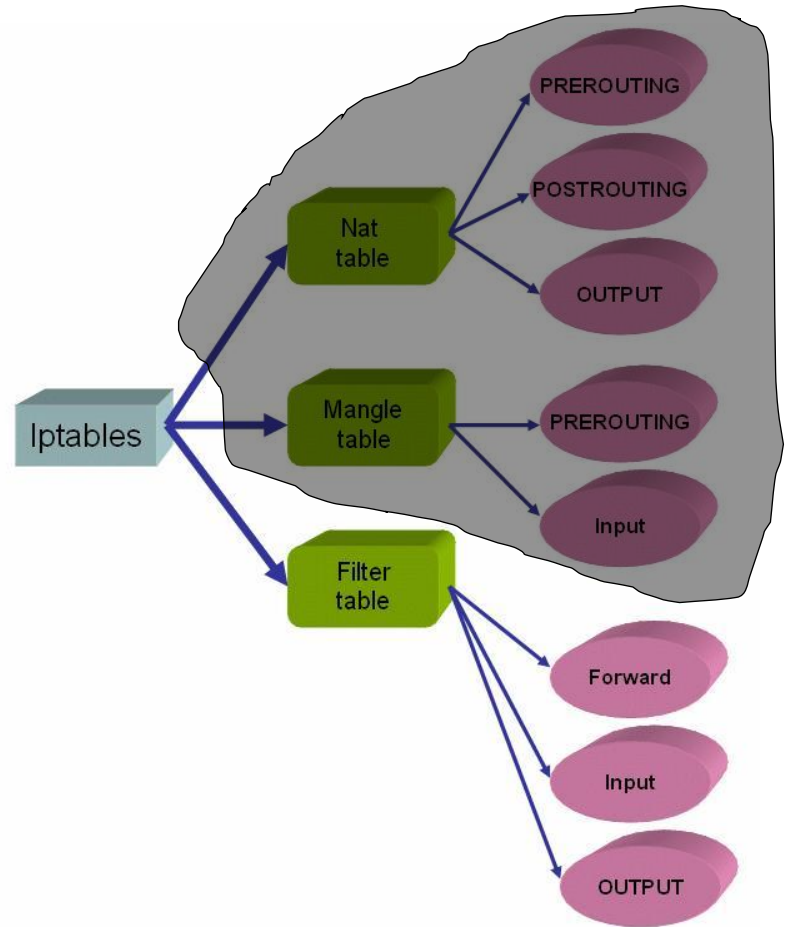
- The same applies to the *postrouting*

- It is allowed to create sequences of rules that can be invoked from within the same basic table



# Firewalls

- `iptables` is the native application on Linux to create and edit rules in all chains (input, output and forward)...
- NOTE: rules created by `iptables` are not persistent:
  - Use `iptables-save` and `iptables-restore` and/or `iptables-apply`; or
  - Use `iptables-persistent` (service)
- For IPv6 use `ip6tables` versions of the same commands



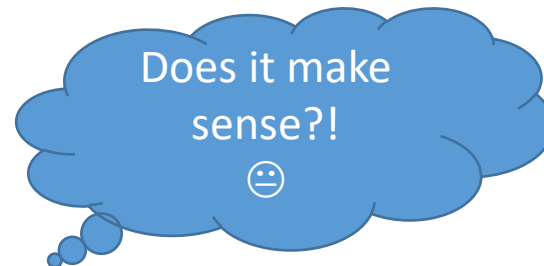
# Firewalls (examples)

```
# iptables -L -v
Chain INPUT (policy DROP 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination
  0      0 ACCEPT    all  --  lo     any     anywhere  anywhere
  0      0 DROP      all  --  any    any     anywhere  anywhere
  0      0 ACCEPT    all  --  any    any     anywhere  anywhere          ctstate RELATED,ESTABLISHED
  0      0 ACCEPT    udp  --  any    any     anywhere  anywhere          udp dpt:ssh
  0      0 ACCEPT    tcp  --  any    any     anywhere  anywhere          tcp dpt:ssh
  0      0 ACCEPT    udp  --  any    any     anywhere  anywhere          udp dpt:http
  0      0 ACCEPT    tcp  --  any    any     anywhere  anywhere          tcp dpt:http

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination

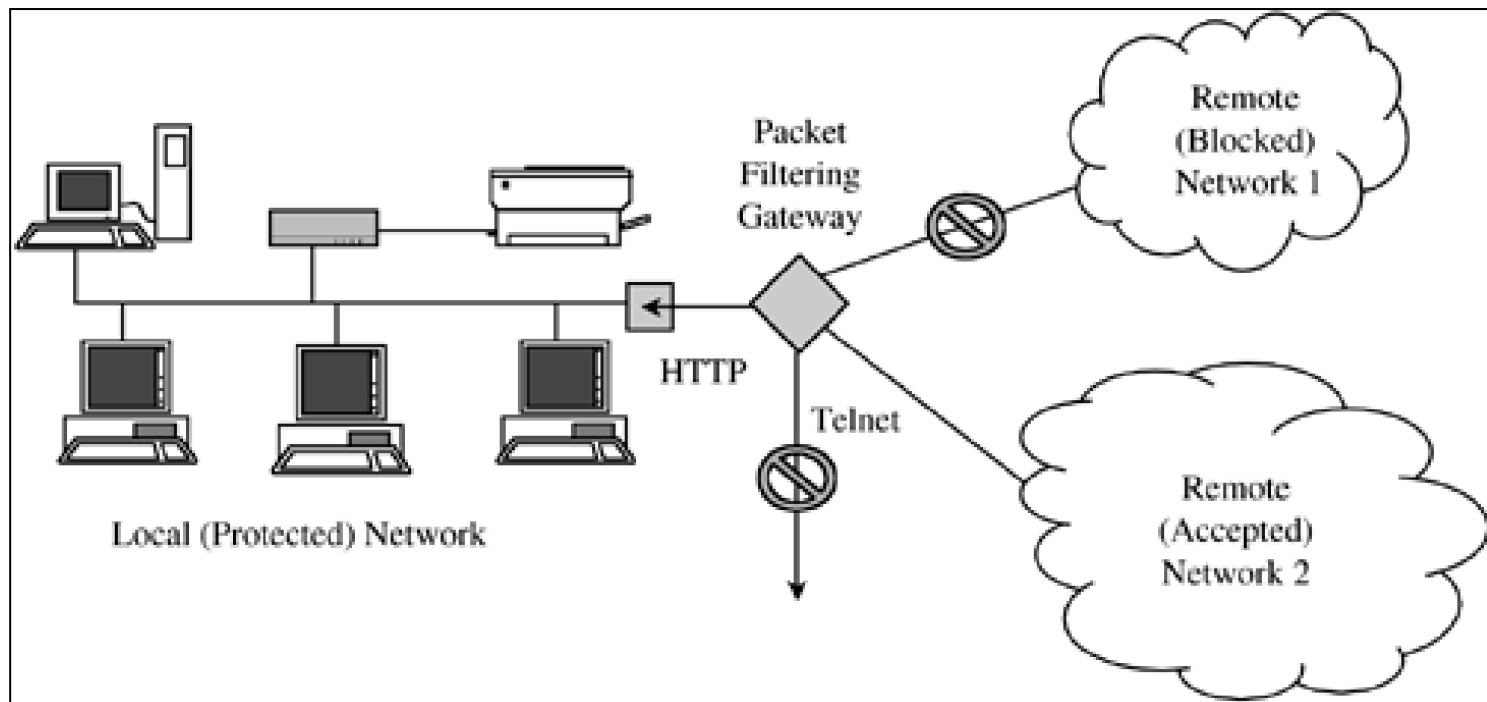
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source    destination
  0      0 ACCEPT    all  --  any    any     anywhere  anywhere
```

```
# iptables -A OUTPUT -p icmp -j REJECT
# iptables -S
-P INPUT DROP
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A INPUT -i lo -j ACCEPT
-A INPUT -f -j DROP
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A INPUT -p udp -m udp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p udp -m udp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A OUTPUT -j ACCEPT
-A OUTPUT -p icmp -j REJECT --reject-with icmp-port-unreachable
```



# Firewalls

- *Packet Filtering Gateway or Screening Routers (2/3)*
  - The simplest solution: filters out addresses and ports (transport level)



# Firewalls

- Clear division between internal and external network
- Must reject external packets with internal source addresses!
  - Don't forget that it is possible to forge addresses
- Major disadvantages
  - Implementation of rules for exceptional cases
  - Maintenance of rules
  - Filters each packet individually



# Firewalls

- *Stateful Inspection Firewall (2/3)*
  - Maintains communication state information between consecutive packets
  - Allows detect attacks based on packet fragmentation
  - More complex rules, allowing to analyze payload
- *Application Proxy Gateway or Bastion Host (4)*
  - Acts as a particular application server (i.e. interprets the protocol)
  - Decides what commands/data will be passed to the actual server and what data is sent out
  - It might even strengthen the authentication function

# Firewalls

- *Guard (4)*

- Typically is a more "intelligent" Bastion Host, with a capacity of (for example):
  - Decide what services request, depending on client features (location, previous uses, etc.)
  - Dynamically limit data transfer
  - Run an anti-virus on embedded files
- Policy definition is a very complex task 😞

- *Personal Firewalls (2/3)*

- Complement the work of (organizational) firewalls "finely" protecting each computer 😊
- Allow Personal Security Policies but it is necessary to manage the rules 😊 😞
- Consume personal machine resources 😞

# Firewalls

| <b><i>Packet Filtering</i></b> | <b><i>Stateful Inspection</i></b>         | <b><i>Application Proxy</i></b>   | <b><i>Guard</i></b>                                | <b><i>Personal Firewall</i></b>                                |
|--------------------------------|-------------------------------------------|-----------------------------------|----------------------------------------------------|----------------------------------------------------------------|
| The simplest                   | Complex                                   | More complex                      | The most complex                                   | Simple                                                         |
| Addresses and ports            | Addresses, ports and data                 | Analyses all package data         | Analyses the entire text of the communication      | Addresses, ports and data                                      |
| Limited audit                  | Audit is possible                         | Auditing of activities            | Auditing of activities                             | Auditing of activities                                         |
| Uses binding rules             | Uses packet sequence information          | Based on the behaviour of proxies | Based on the interpretation of the message content | Uses binding rules (typically)                                 |
| Complex configuration (rules)  | Normally configured to stop known attacks | Less rules but more powerful      | Very complex functionality can limit security!     | Typically dynamic configuration (starts by denying everything) |

# Firewalls

- Limitations

- Only secure if the firewall filters all the network perimeter
- Information that comes out (legitimately) the perimeter is not secure
- **The configuration must be carefully planned and managed**
  - Some recent developments: “topoS”, a formal process for synthesis of secure network configurations  
Diekmann C, Korsten A, Carle G (2015) Demonstrating topoS: Theorem-prover-based synthesis of secure network configurations. In: 2015 11th International Conference on Network and Service Management (CNSM). pp 366–371  
[Project homepage](#)

# Firewalls

- Limitations

- Is the most visible network component and therefore more subject to attacks (its defence should be taken carefully)
- Malicious data, concealed within legitimate communications, cannot be avoided – defences at the application level...
- Limited capacity with encrypted traffic

# Firewalls

Figure 1. Magic Quadrant for Network Firewalls



Source: Gartner (November 2020)

# Firewalls

- UTM (Unified Threat Management) or NGFW (Next Generation Firewall) ? Additional

| Resource →<br>↓Product    | FW/VPN | IPS | AV  | Web<br>Filtering | Application<br>Detection | Email<br>Security | DLP |
|---------------------------|--------|-----|-----|------------------|--------------------------|-------------------|-----|
| Next Generation Firewalls |        |     |     |                  |                          |                   |     |
| Checkpoint                | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| McAfee                    | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| Palo Alto Networks        | Sim    | Sim | Sim | Sim              | Sim                      | ???               | Sim |
| Sourcefire                | Sim    | Sim | Sim | Sim              | Sim                      | ???               | Sim |
| UTM                       |        |     |     |                  |                          |                   |     |
| Astaro                    | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| Fortinet                  | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| Sonicwall                 | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| Watchguard                | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Sim |
| Additional                |        |     |     |                  |                          |                   |     |
| Cisco                     | Sim    | Sim | Sim | Sim              | Não                      | Sim               | Não |
| Juniper                   | Sim    | Sim | Sim | Sim              | Sim                      | Sim               | Não |

<http://www.itfriends.org/ngfw-vs-utm/>

# Notes and reflections



- **Thomason, S. "Improving Network Security: Next Generation Firewalls and Advanced Packet Inspection Devices." Global Journal of Computer Science and Technology 12.13-E (2012).**



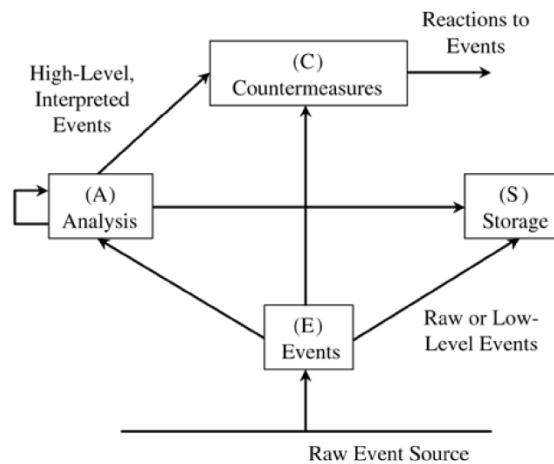
- **Are Tomorrow's Firewalls Finally Here Today?**



# Intrusion Detection Systems (IDS)

- *Firewalls* and Access Control prevent malicious activity
- What about bad “legitimate” utilization? It must be detected ...  $\Rightarrow$  IDS
  - Typically, a computer system looking for “suspected” activity

Base IDS  
Architecture



- Monitors the activity of users and machines
- Audit systems configuration and vulnerabilities
- Evaluation of the integrity of critical files
- Identification of known attacks
- Identification of abnormal activity
- Identification of configuration errors
- Management of intrusion traps
- ...

# Intrusion Detection Systems (IDS)

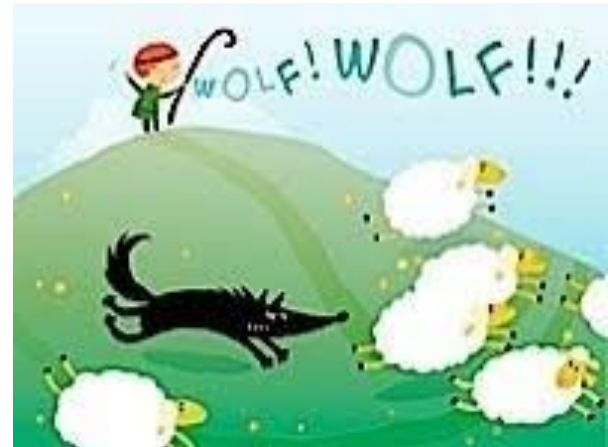
- Relevant properties

- Efficiency

- Detection problem: false positives

- Extensibility

- Adaptability



- Typically, an IDS:

- performs in **stealth mode** (e.g., two network interfaces)
  - requires a great deal of human intervention ☹️

# Intrusion Detection Systems (IDS)

- IDS types (**operation based classification**)
  - Signature-based (SB)
    - Verification of patterns associated with **known attacks**  
Suricata; SNORT; IDIOT; STAT; ...
  - Anomaly-based (AB) – Heuristics
    - Build a model of **normal or acceptable behaviour** and signals everything that does not check this model  
Zeek (Bro); IDES
- IDS types (**characteristic function**)
  - Targeted to networks (**NIDS**)
    - A single component that monitors network traffic
  - Targeted to computers (**HIDS**)
    - Run on a computer, “observing” their behaviour (log analysers; file integrity checkers;...)

# Intrusion Detection Systems (IDS)

|      | Advantages                                                                                                                                                                                                            | Limitations                                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NIDS | <ul style="list-style-type: none"><li>• Monitors multiple nodes</li><li>• No impact on the operation of the network</li><li>• Undetectable (if properly configured)</li><li>• Pre-attack activity detection</li></ul> | <ul style="list-style-type: none"><li>• Traffic volume</li><li>• Incapable to detect if an attack was successful</li><li>• Hardly sensing fragmented attacks (by various packets)</li><li>• Do not analyse encrypted traffic</li></ul>                                                                        |
| HIDS | <ul style="list-style-type: none"><li>• Effective to detect the consequence of attacks</li><li>• Ability to detect attacks based on cipher techniques</li></ul>                                                       | <ul style="list-style-type: none"><li>• Demanding administration (one per node)</li><li>• More exposed, by operating in target machines; more easily annulled</li><li>• Inadequate to detect the early stages of attacks (e.g., network scanning)</li><li>• Consume resources of monitored machines</li></ul> |

# Intrusion Detection Systems (IDS)

- Classification according to the system characteristics
  - Time-to-detection: real-time vs batch
  - Granularity of data: typically, real-time  $\Rightarrow$  smaller amounts of data; batch  $\Rightarrow$  larger amounts of data
  - Locus of data-processing: central vs. distributed processing
  - Locus of data-gathering: locally vs remotely collected data
  - Alarm response
    - Seeks to collect all related data
    - Protects the resource under threat - **Active**
    - "Calls" an operator - **Passive**

# Network IDS

- Typical architecture of a hybrid NIDS

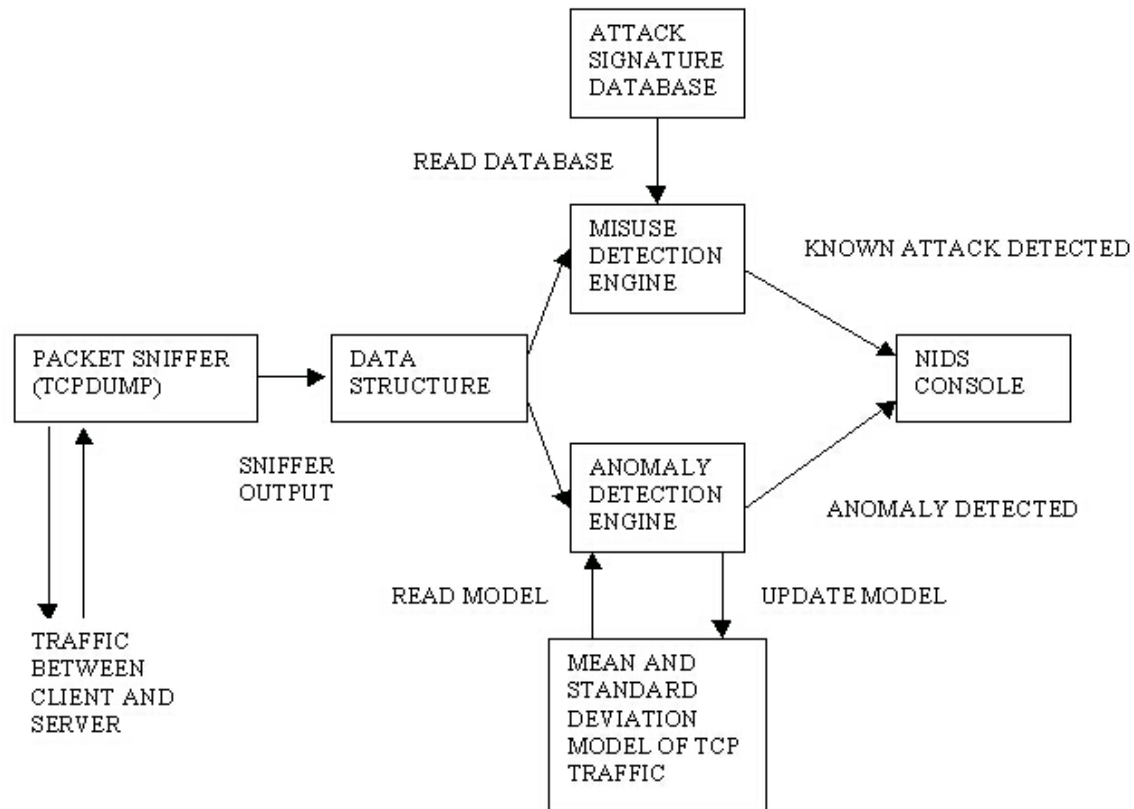
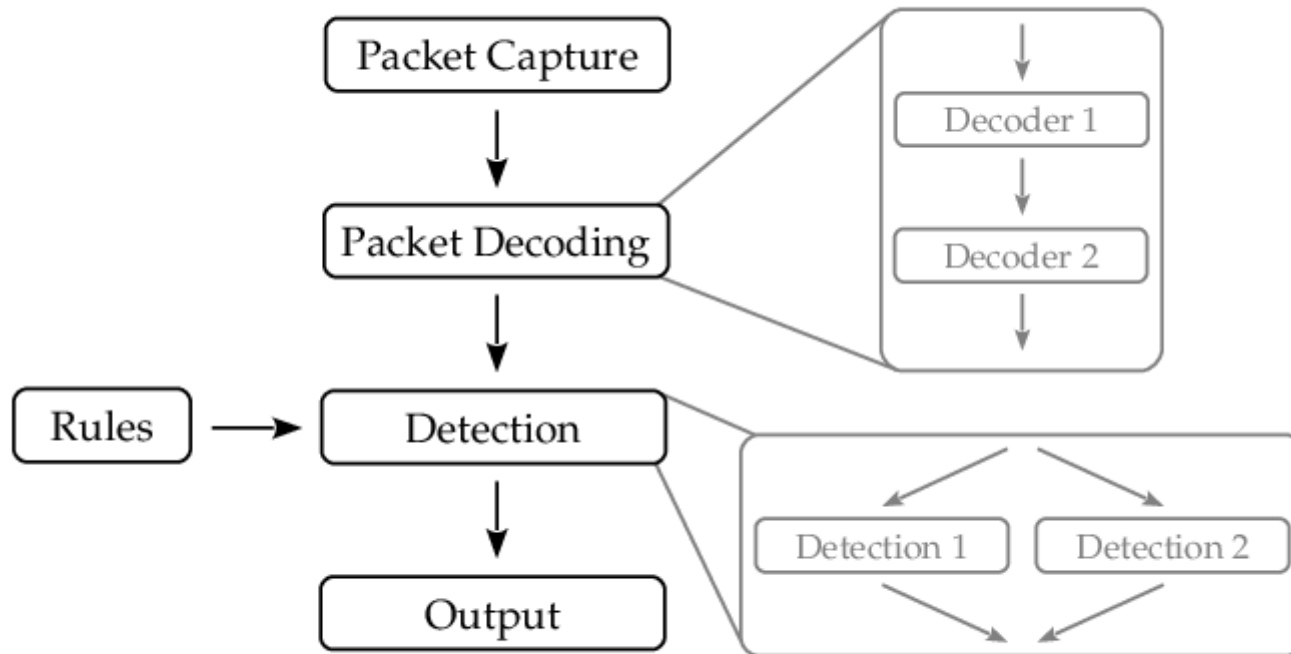


Figure 1: NIDS Architecture

# Network IDS

- What role is typically assigned to NIDS?
  - Vulnerability identification
  - Network security auditing
  - Check security policies violation
  - Detect attacks initiated by internal systems!
  - Support incident response and forensic analysis
  - Complement other defence mechanisms 😊

# Suricata Architecture



[https://www.researchgate.net/figure/Suricata-architecture\\_fig3\\_309229246](https://www.researchgate.net/figure/Suricata-architecture_fig3_309229246)



# Intrusion Detection Systems (IDS)

- Classification of detection techniques

|                    |               |                             |                  |                                                       |
|--------------------|---------------|-----------------------------|------------------|-------------------------------------------------------|
| signature          | programmed    | State-modelling             | State-transition | USTAT                                                 |
|                    |               |                             | Petri-net        | IDIOT                                                 |
|                    |               | Expert-system               |                  | <b>NIDES; EMERALD; MIDAS-direct; DIDS; MIDAS(2)</b>   |
|                    |               | String-matching             |                  | NSM; <b>SNORT</b> ; <b>SURICATA</b>                   |
|                    |               | Simple rule-based           |                  | <b>NADIR; NADIR(2); ASAX; Bro; JiNao; Haystack(2)</b> |
| Signature inspired | Self-learning | Automatic feature selection |                  | Ripper                                                |

# Intrusion Detection Systems (IDS)

- Classification of detection techniques

|         |               |                        |                              |                                               |
|---------|---------------|------------------------|------------------------------|-----------------------------------------------|
| Anomaly | Self-learning | Non time series        | Rule modelling               | WebStalker                                    |
|         |               |                        | Descriptive statistics       | <b>IDEAS; NIDES; EMERALD; JiNao; Haystack</b> |
|         |               | Time series            | ANN (Artificial Neural Net.) | Hyperview                                     |
|         | programmed    | Descriptive statistics | Simple statistic             | <b>MIDAS; NADIR; Haystack</b>                 |
|         |               |                        | Simple rule-based            | NSM                                           |
|         |               |                        | Threshold                    | ComputerWatch                                 |
|         |               | Default deny           | State series modelling       | <b>DPEM; JANUS; (Zeek) Bro</b>                |

# Intrusion Prevention Systems (IPS)

Figure 1. Magic Quadrant for Intrusion Detection and Prevention Systems



Source: Gartner (January 2018)

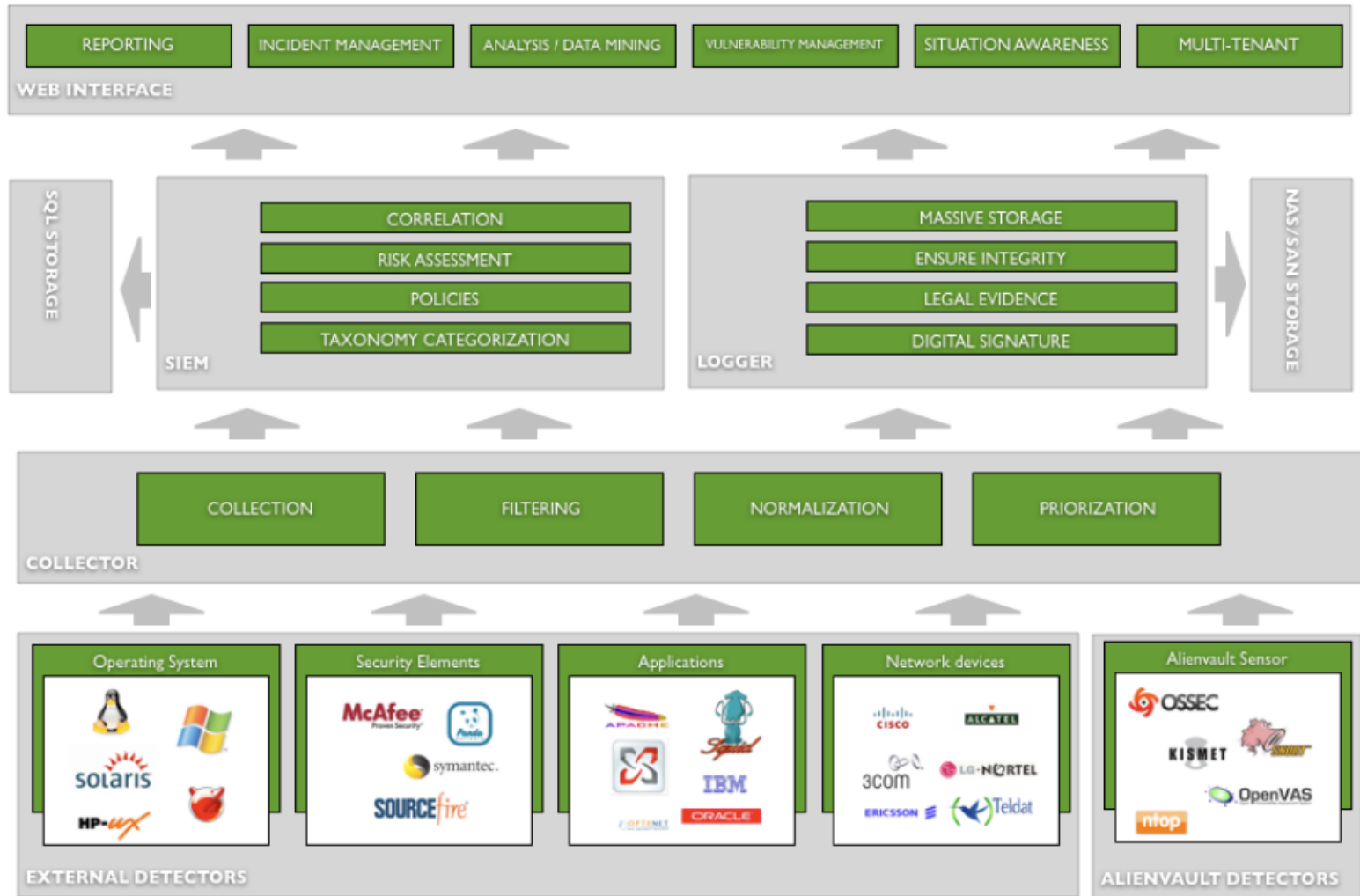
# SIEM (SecInfo Event Manager)

- Framework for intrusion detection and event management. Example: OSSIM
  - NIDS: Snort
  - HIDS: Osiris; OSSEC; Snare
  - Vulnerability analysis: Nessus
  - Research and inventory: Nmap; OCS
  - Anomaly detection: Spade; p0f; Pads; Arpwatch
  - Network monitor: Ntop; Tcptrack; Nagios
  - Logs monitor: Ntsyslog; Syslog; Snarewindows
  - System information gathering: Cisco PIX; IPTables; IIS Collector; Apache Collector
  - Antivirus: ClamAv

# OSSIM

- 3-tier architecture:
  - Frontend: web interface for visualization
  - Server and *Frameworkd*: standardisation, prioritization, correlation, risk evaluation and storage (SQL)
  - Detectors: modules for capture and monitoring security events (individually highlight limitations concerning sensitivity – False Negatives – and reliability – False Positives)

# OSSIM



# Security Onion

- “Security Onion is a Linux distro for intrusion detection, network security monitoring, and log management.”
  - Snort
  - Suricata
  - Zeek (Bro)
  - OSSEC
  - Sguil
  - Squert
  - Snorby
  - ELSA
  - Xplico
  - NetworkMiner
  - ...

<http://blog.securityonion.net/p/securityonion.html>

# SIEM (SecInfo Event Manager)



Source: Gartner (February 2020)



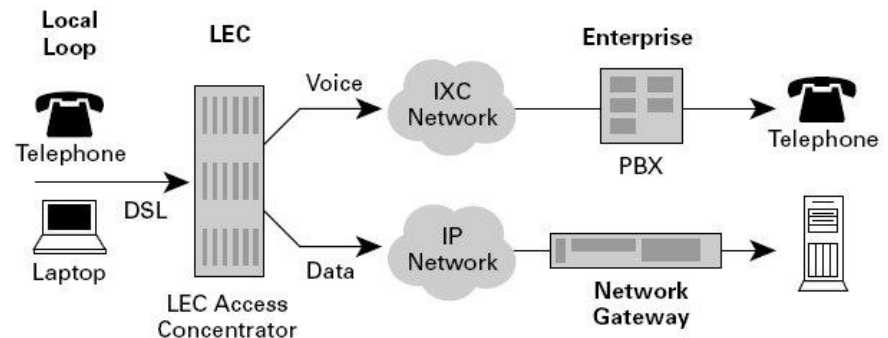
# Notes and reflections

- González-Granadillo, Gustavo, Susana González-Zarzosa, and Rodrigo Diaz. "[Security information and event management \(SIEM\): analysis, trends, and usage in critical infrastructures.](#)" Sensors 21.14 (2021): 4759.
- Kavanagh, Kelly M., Oliver Rochford, and Toby Bussa. "[Magic quadrant for security information and event management.](#)" Gartner Group Research Note (2015).

# Notes about VPN

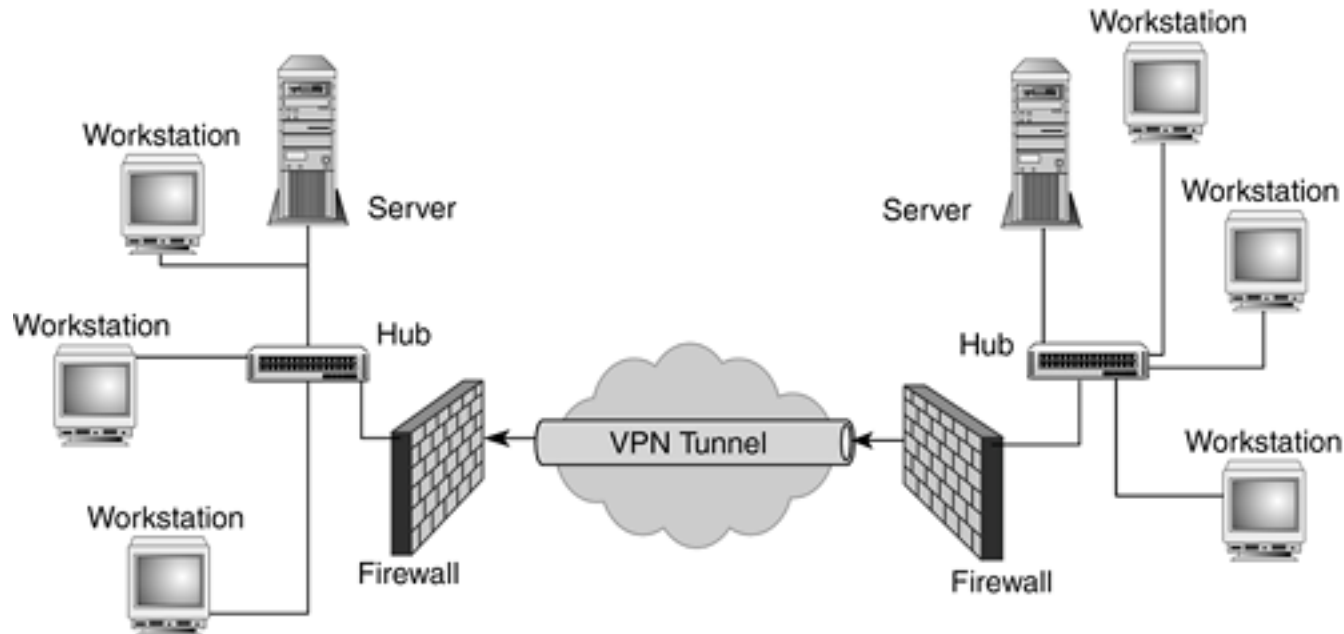
- Secure remote access

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  - Economic factors
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  - Bandwidth
  - Any "last mile" technology



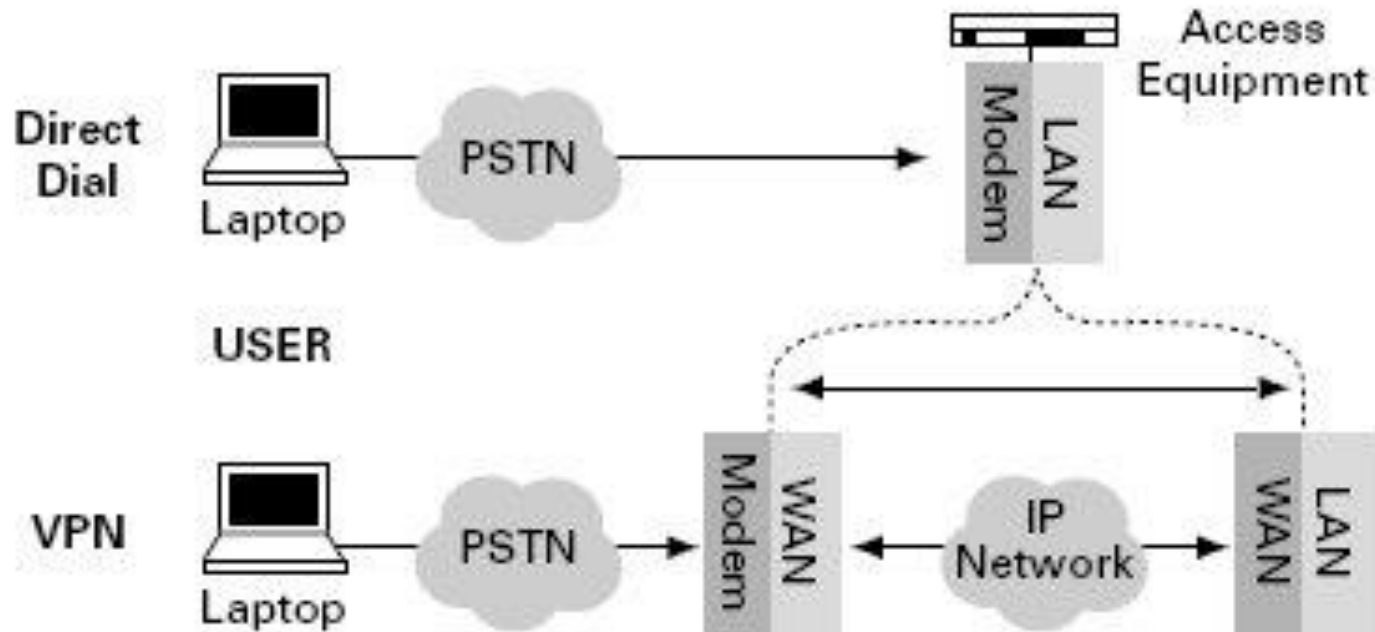
# Notes about VPN

- VPN (Virtual Private Network) concerning DoD compensates the only downside of the DoD's open philosophy, **security**
- VPN uses the Internet to establish a virtual circuit (or tunnel), to interconnect two terminals



# Notes about VPN

- Is (still) possible a hybrid solution (DoV until the access point-ISP)



# Notes about VPN

- Protocols
  - Typically, in three phases:
    - Client authentication on a firewall
    - Firewall responds with a session key
    - Client and server communicate on a secure channel encrypted with this key
  - PPTP (Point-to-Point Tunnel Protocol), is an extension to PPP and enables to encapsulate TCP/IP, IPX/SPX and NetBEUI. PPP packets can be encrypted using MPPE (Microsoft Point-to-Point Encryption – 128 bits, starting with SP2 of W2k). Authentication can be done using MS-CHAP, MS-CHAPv2 or EAP/TLS – **breakable (NSA)**
  - L2TP (Layer 2 forwarding plus PPTP); uses IPSec to data encryption, greatly increasing the security level => Server and client support L2TP and IPSec (This is not the case of **Windows NT** and **Windows 9x**)

# Notes about VPN

- Steps for implementing a VPN
  - Number of ports (or number of simultaneous virtual sessions) – typical 1 to 10 users
  - The available bandwidth (WAN + server) will be spread by all ports =>  $LB_{min}/port$
  - Classify users:
    - Remote offices
    - Mobility scheme
    - Telework scheme
    - Time extension scheme
  - Authentication:
    - Username/password
    - **Tokens**
    - **Digital certificates**
  - Dedicated hardware (Cisco, HP, Intel, multiple routers)
  - Native support in Linux and Windows (2K, XP, ... ☺), and several other solutions

# Notes about VPN

- Security limitations (particularly when using third-party services)
  - Logging Policies (trustable third party?)
    - Is it possible to negotiate a no-log policy?
    - VPN provider reputation
  - Authentication flaws
  - Data leaks
    - [DNS, IP](#)
    - [WebRTC \(Web Real-Time Communication\)](#)

# Notes about VPN

- Security limitations (cont)
  - **Free VPN providers** are usually business related to advertisers (Privacy issues); further, they can abuse your IP and bandwidth
    - ProtonVPN
    - Surfshark VPN
    - ...
  - Poorly configured encryption
    - PPTP braked by NSA some time ago
    - SSL/TLS vulnerable implementations
  - End point security-level (malware)
  - Single point of failure



# Notes about VPN

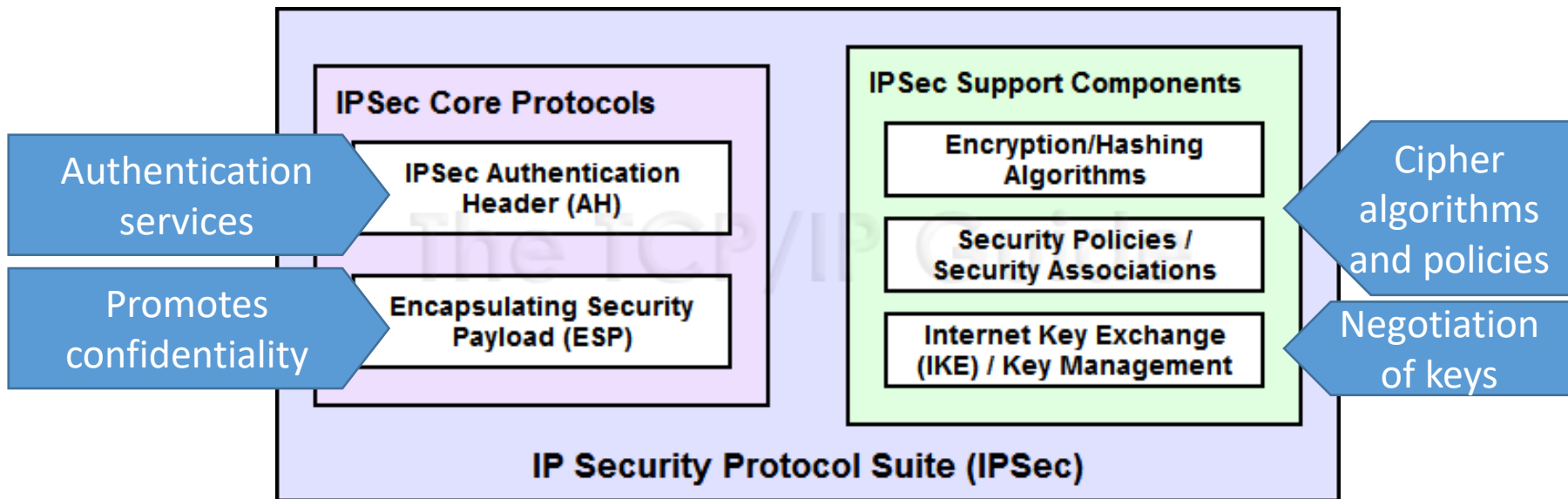
- Platforms (Open Source)

- [OpenVPN](#) – a complex cross-platform software (very reliable, state-of-the-art security protocols)
  - Supports most protocols and Access Control mechanisms; very flexible, but very **hard to configure** correctly
  - Two versions: Community Edition – free, but needs a deep understanding of configuration details (some scripts available, like [this one](#)), no GUI for server administration; Access Server – GUI based configuration with support, but with a paid licence based on the number of connections (free up to 3 connections); an interesting comparison is [available here](#)
- [Wireguard](#) – based on OpenSSL and also cross-platform
  - Enforces security using advanced cryptographic algorithms
  - Node authentication using public keys for tunnelling protocol – *Cryptokey Routing*
  - No GUI for server administration (yet)
- [strongSwan](#) – based on IPSec and also cross-platform
  - No GUI for server administration

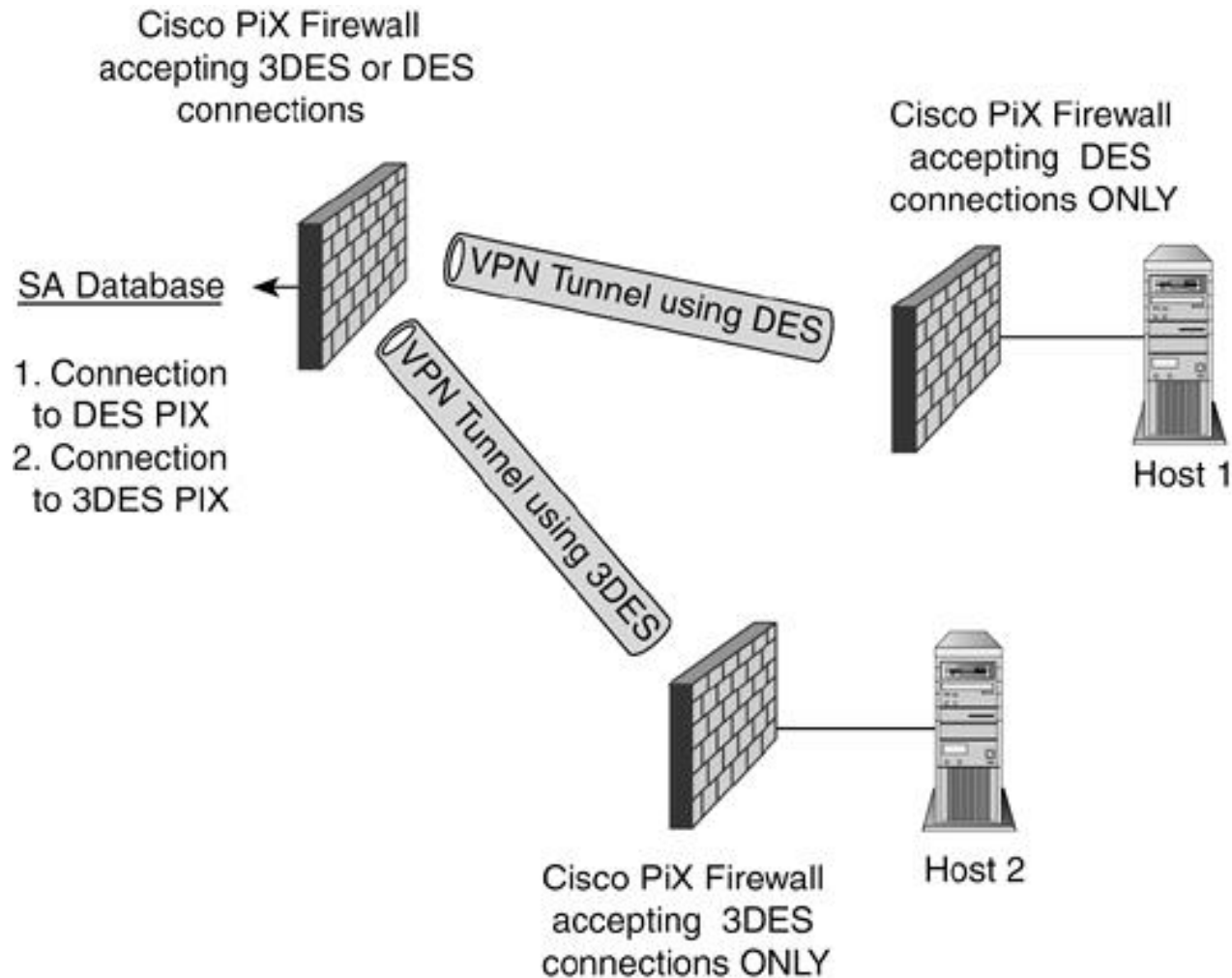
# Notes about IPSec

- Promotes: **confidentiality, integrity** and **authenticity**
- Standard defined by the IETF for secure communication ("real-time")
- Assumes that the parties share a session key (distributed manually, or with IKE-below)
- IPSec works with both IPv4 and IPv6
- **SA (Security Association)** denotes an **unidirectional connection cryptographically secure** (address, sequence numbers, encryption algorithms, ensures or not integrity, etc.). Each SA has a key and a unique identification, SPI – Security Parameter Index
- The IPSec header identifies the respective SA
- Any system that implements IPSec has to have a database of SAs (**SAD** – Security Association Database)
- Each host must be configured with all necessary parameters, which are stored in a SPD – Security Policy Database

# IPSec – main protocols

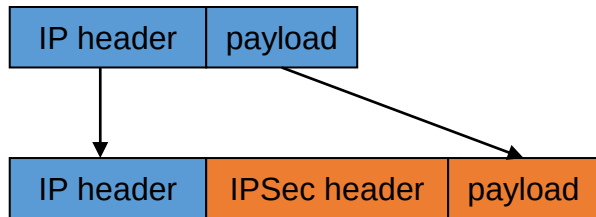


# Notes about IPSec

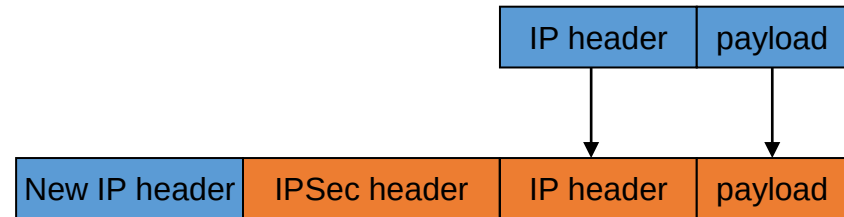


# Notes about IPSec

Transport Mode



Tunnel Mode



- The IPSec packet header can have two distinct fields:
  - AH (Authentication Header) – ensures integrity (also concerning parts of the IP header) and sender/IP authenticity
  - ESP (Encapsulating Security Payload) – ensures confidentiality and/or integrity (concerning packet content) and sender/IP authenticity
- Tunnel Mode: Host-to-Host; Host-to-Gateway; **Gateway-to-Gateway**

# Notes about IPSec

- IPSec is an integral part of IPv6
- Tunnel mode creates problems with NAT (Network Address Translation) and Firewall, preventing them from analyse/change IP headers!
- The AH and ESP headers (**attributes for each SA**) are indicated in the IP header by Protocol Numbers 51 and 50, respectively.

# Notes about IPSec

## Header IPv4

|         |                              |
|---------|------------------------------|
| 4 bits  | Version                      |
| 4 bits  | Header size (32-bit units)   |
| 8 bits  | Type of service              |
| 16 bits | Size of header + data        |
| 16 bits | Package identification       |
| 3 bits  | Flags                        |
| 13 bits | Fragment offset              |
| 8 bits  | Hops (or Time To Live - TTL) |
| 8 bits  | <b>PROTOCOL</b>              |
| 16 bits | Header checksum              |
| 32 bits | Source address               |
| 32 bits | Destination address          |
| Var     | Options                      |

## Header IPv6

|          |                                           |
|----------|-------------------------------------------|
| 4 bytes  | <b>Version (4 bits)   Type of service</b> |
| 2 bytes  | Packet size                               |
| 1 byte   | <b>NEXT HEADER</b>                        |
| 1 byte   | Hops (or Time To Live - TTL)              |
| 16 bytes | Source address                            |
| 16 bytes | Destination address                       |

## Header extensions

|        |                     |
|--------|---------------------|
| 1 byte | Next header         |
| 1 byte | Header size         |
| Var    | Data in this header |

# Notes about IPSec

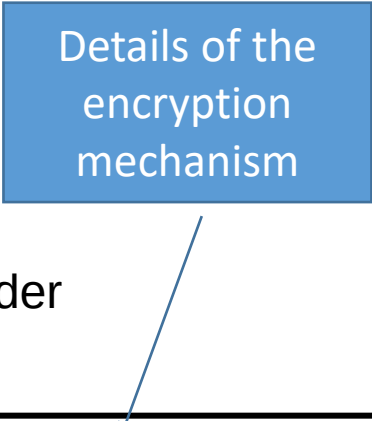
## AH Header

|         |                                       |
|---------|---------------------------------------|
| 1 byte  | Next header                           |
| 1 byte  | Packet size                           |
| 2 bytes | Unused                                |
| 4 bytes | <b>SPI (Security Parameter Index)</b> |
| 4 bytes | Sequence number                       |
| Var     | <b>Authentication data</b>            |

## ESP Header

|         |                                       |
|---------|---------------------------------------|
| 4 bytes | <b>SPI (Security Parameter Index)</b> |
| 4 bytes | Sequence number                       |
| Var     | <b>IV (Initialization Vector)</b>     |
| Var     | Data                                  |
| Var     | Padding                               |
| 1 byte  | Padding size (in bytes)               |
| 1 byte  | Next header/protocol type             |
| Var     | <b>Authentication data</b>            |

Details of the  
encryption  
mechanism





# Notes about IKE

- IKE (Internet Key Exchange) – Mutual authentication, based on a **prolonged use shared key**. Used to deduce a session key for each SA
- IPSec can be configured without IKE, but loses security and flexibility
- Two protocols:
  - ISAKMP (**Internet Security Association and Key Management Protocol**) – a generic framework to establish, negotiate, modify and eliminate SAs.
  - Oakley (Variant of Diffie-Hellman) and Skeme Key Exchange (generate and exchange symmetric keys)

# Notes about IKE

- Oakley (based on the Diffie-Hellman algorithm)
  - **A** and **B** choose two prime numbers  $n$  and  $g$  such that  $g$  is primitive mod  $n$ ;  $(n-1)/2$  must also be a prime number. These numbers are exchanged on a non secure way.
  - **A** chooses a random number  $x$  and calculates  $X = g^x \bmod n$   
Sends  $X$  to **B**
  - **B** chooses a random number  $y$  and calculates  $Y = g^y \bmod n$   
Sends  $Y$  to **A**
  - **A** calculates  $\mathbf{K} = Y^x \bmod n$
  - **B** calculates  $\mathbf{K}' = X^y \bmod n$
  - By definition  $\mathbf{K} \equiv \mathbf{K}' = g^{xy} \bmod n$  and will be the session key
  - There are small variants of this algorithm even more robust

# Notes about IKE

- IKE – Phase 1

- Remote user authentication
  - Shared keys or certificates previously distributed (the most common) – embedded in applications
  - Two modes: **normal**; and **aggressive** (faster but less secure)
- Exchange of the public key that will be used to encrypt the exchange of information on stage 2

- IKE – Phase 2

- Exchange of configuration information (encrypted)
- Only one mode (**quick mode**), similar to phase 1 aggressive mode

# Notes about IKE

- Example of traffic with a Cisco VPN server (IKEv1)

| No. - | Time            | Source        | Destination   | Protocol | Info                      |         |
|-------|-----------------|---------------|---------------|----------|---------------------------|---------|
| 5     | 02:15:38.403042 | 192.168.1.100 | 193.137.17.1  | UDP      | Source port: 15           | Phase 1 |
| 6     | 02:15:38.412247 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Aggressive                |         |
| 7     | 02:15:38.613099 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Aggressive                |         |
| 8     | 02:15:38.621266 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Aggressive                |         |
| 9     | 02:15:38.622075 | 192.168.1.100 | 193.137.17.1  | UDPENCAP |                           |         |
| 10    | 02:15:38.681015 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Transaction (Config Mode) | Phase 2 |
| 13    | 02:15:47.745493 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Transaction (Config Mode) |         |
| 14    | 02:15:48.103899 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Transaction (Config Mode) |         |
| 15    | 02:15:48.147820 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Transaction (Config Mode) |         |
| 16    | 02:15:48.264967 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Quick Mode                |         |
| 17    | 02:15:48.331755 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Quick Mode                |         |
| 19    | 02:15:49.015304 | 192.168.1.100 | 193.137.17.1  | UDP      | Source port: 62           |         |
| 20    | 02:15:49.302615 | 192.168.1.100 | 193.137.17.1  | UDPENCAP |                           |         |
| 21    | 02:15:49.389736 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Transaction (Config Mode) |         |
| 22    | 02:15:49.390301 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Transaction (Config Mode) |         |
| 28    | 02:15:58.446974 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Informational             |         |
| 29    | 02:15:58.517844 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Informational             |         |
| 30    | 02:16:08.993792 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Informational             |         |
| 31    | 02:16:09.055528 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Informational             |         |
| 33    | 02:16:39.447423 | 192.168.1.100 | 193.137.17.1  | UDPENCAP |                           |         |
| 34    | 02:16:49.449768 | 192.168.1.100 | 193.137.17.1  | UDPENCAP |                           |         |
| 35    | 02:16:53.081483 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Informational             |         |
| 36    | 02:16:53.082895 | 192.168.1.100 | 193.137.17.1  | ISAKMP   | Informational             |         |
| 37    | 02:16:53.146336 | 193.137.17.1  | 192.168.1.100 | ISAKMP   | Informational             |         |

# Notes about IKE

- Example of traffic with a Cisco VPN server (IKEv1)

| Time       | Source        | Destination   | Protocol | Length | Info                      |                                                                           |
|------------|---------------|---------------|----------|--------|---------------------------|---------------------------------------------------------------------------|
| 0.00000... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 1343   | Aggressive                | Phase 1:<br>- Policy negotiation<br>- DH key exchange<br>- Authentication |
| 0.02311... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 486    | Aggressive                |                                                                           |
| 0.02653... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 218    | Aggressive                |                                                                           |
| 0.04799... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 122    | Transaction (Config Mode) |                                                                           |
| 0.04817... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 154    | Transaction (Config Mode) |                                                                           |
| 0.10054... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 122    | Transaction (Config Mode) |                                                                           |
| 0.10101... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 122    | Transaction (Config Mode) |                                                                           |
| 0.10125... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 234    | Transaction (Config Mode) |                                                                           |
| 0.12110... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 586    | Transaction (Config Mode) | Phase 2:<br>- Establish security parameters for protected traffic         |
| 0.16498... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 666    | Quick Mode                |                                                                           |
| 0.18503... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 138    | Informational             |                                                                           |
| 0.18555... | 193.137.17.1  | 192.168.1.126 | ISAKMP   | 250    | Quick Mode                |                                                                           |
| 0.18605... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 106    | Quick Mode                |                                                                           |
| 0.18630... | 192.168.1.126 | 193.137.17.1  | ISAKMP   | 138    | Informational             |                                                                           |
| 0.18634... | 192.168.1.126 | 193.137.17.1  | ESP      | 142    | ESP (SPI=0xa4999070)      |                                                                           |
| 0.18636... | 192.168.1.126 | 193.137.17.1  | ESP      | 142    | ESP (SPI=0xa4999070)      |                                                                           |

# Notes about IPSec

- Principais fragilidades
  - Eventual limitação no espaço de endereçamento IP (IPv4)
  - Falta de um mecanismo autónomo de configuração
  - Falta de segurança intrínseca!
    - Facilidade na identificação do software (*fingerprint*)
    - Armazenamento das credenciais do utilizador, no cliente
    - Vulnerabilidades no modo agressivo com *Pre Shared Key* (normalmente a password, que pode ser deduzida). Isto facilita um ataque do tipo MITM.
    - Ausência de um mecanismo do tipo *Account Lockout* – em função do número de tentativas (permite um ataque de força bruta)
    - Configurações por defeito fracas; documentação fraca
- Fragilidades herdadas do IP, pensado para ambientes de aplicação mais reduzidos (DoS...)
- O IPv6 resolve muitas das questões, mas deixa outras por resolver...

# Exercise

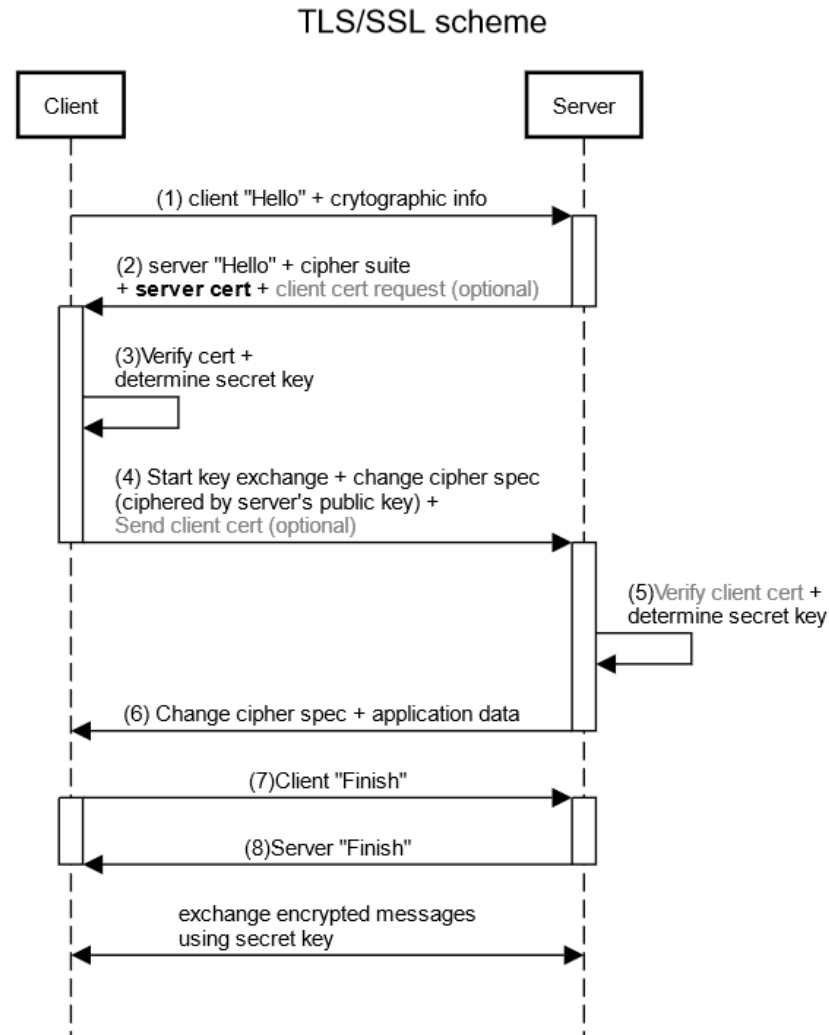
- Much has been discussed about the Elimination of TCP/IP vulnerabilities when IPv6 protocol is adopted. Is it so?

# Notes about TLS/SSL

- SSL (Secure Sockets Layer) – Protocol created by Netscape
- Positioned between application and transport levels (TCP), and ensures:
  - Server authentication
  - Optional client authentication (more versatile in TLS, using TLS-PSK, a shared key mechanism)
  - Encrypted channel with shared key (negotiated algorithm – 3DES, RC4, etc); **symmetric key generation algorithm more resistant in the TLS**
- Protocol summary:
  - Client requests an SSL session
  - Server responds with its digital certificate (public key)
  - The “**Cipher Suite**” is negotiated
  - Client sends a seed for the session key, encrypted with the server’s public key, and they both calculate the session key
  - Optionally the messages are authenticated (hash function – HMAC – in TLS; pseudo random function in SSL)



# Notes about TLS/SSL



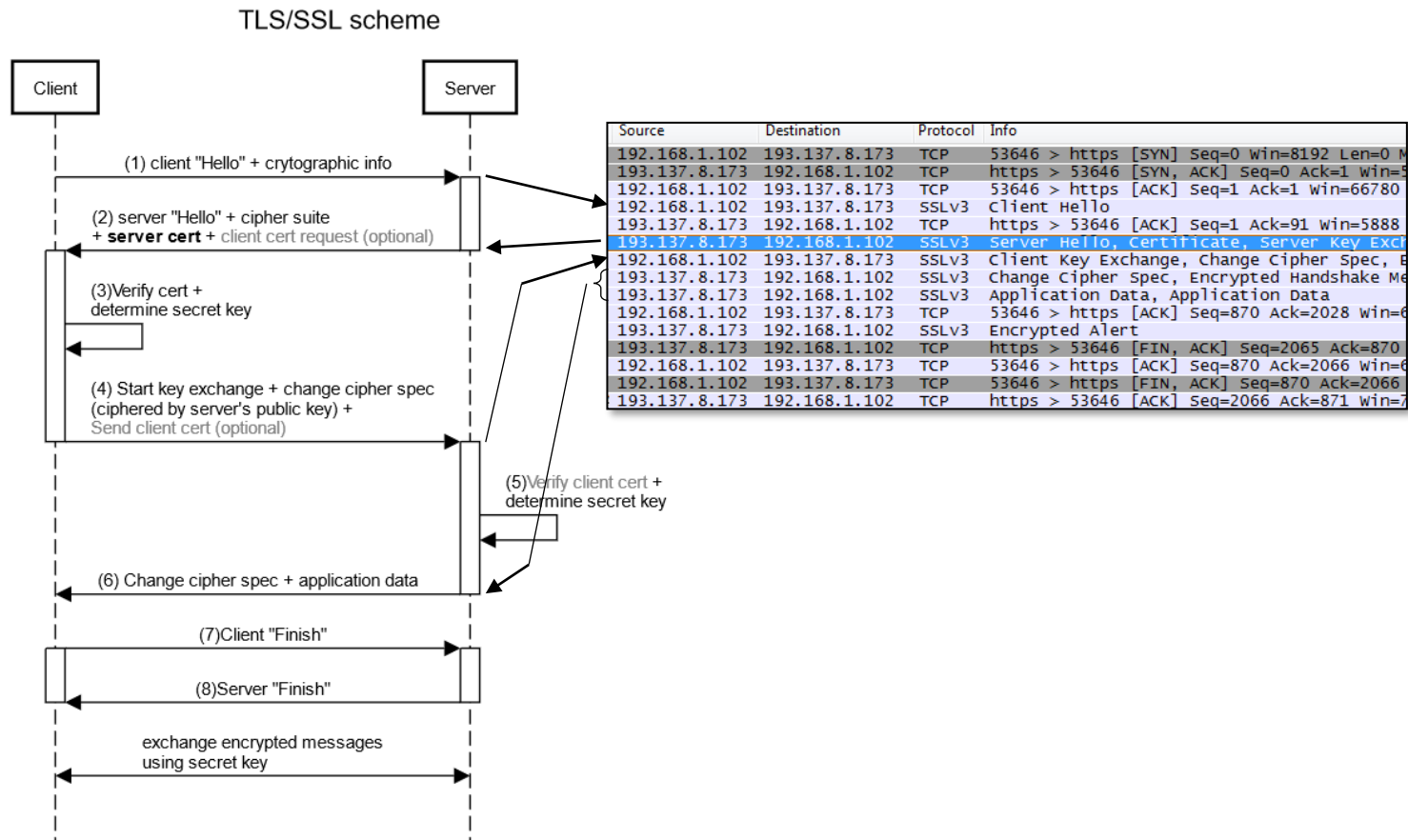
# Notes about TLS/SSL

| No. | Time      | Source        | Destination   | Protocol | Info                                                                 |
|-----|-----------|---------------|---------------|----------|----------------------------------------------------------------------|
| 56  | 2.076333  | 192.168.1.102 | 193.137.8.173 | TCP      | 53646 > https [SYN] Seq=0 win=8192 Len=0 MSS=1260 WS=2               |
| 57  | 2.103351  | 193.137.8.173 | 192.168.1.102 | TCP      | https > 53646 [SYN, ACK] Seq=0 Ack=1 win=5840 Len=0 MSS=1460 WS=6    |
| 58  | 2.103409  | 192.168.1.102 | 193.137.8.173 | TCP      | 53646 > https [ACK] Seq=1 Ack=1 win=66780 Len=0                      |
| 59  | 2.103739  | 192.168.1.102 | 193.137.8.173 | SSLv3    | Client Hello                                                         |
| 60  | 2.151590  | 193.137.8.173 | 192.168.1.102 | TCP      | https > 53646 [ACK] Seq=1 Ack=91 win=5888 Len=0                      |
| 61  | 2.166328  | 193.137.8.173 | 192.168.1.102 | SSLv3    | Server Hello, Certificate, Server Key Exchange, Server Hello Done    |
| 62  | 2.168825  | 192.168.1.102 | 193.137.8.173 | SSLv3    | Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message |
| 63  | 2.208830  | 193.137.8.173 | 192.168.1.102 | SSLv3    | Change Cipher Spec, Encrypted Handshake Message                      |
| 64  | 2.210316  | 193.137.8.173 | 192.168.1.102 | SSLv3    | Application Data, Application Data                                   |
| 65  | 2.210346  | 192.168.1.102 | 193.137.8.173 | TCP      | 53646 > https [ACK] Seq=870 Ack=2028 win=66780 Len=0                 |
| 66  | 2.210370  | 193.137.8.173 | 192.168.1.102 | SSLv3    | Encrypted Alert                                                      |
| 67  | 2.210512  | 193.137.8.173 | 192.168.1.102 | TCP      | https > 53646 [FIN, ACK] Seq=2065 Ack=870 win=7424 Len=0             |
| 68  | 2.210532  | 192.168.1.102 | 193.137.8.173 | TCP      | 53646 > https [ACK] Seq=870 Ack=2066 win=66740 Len=0                 |
| 77  | 9.990855  | 192.168.1.102 | 193.137.8.173 | TCP      | 53646 > https [FIN, ACK] Seq=870 Ack=2066 win=66740 Len=0            |
| 79  | 10.019784 | 193.137.8.173 | 192.168.1.102 | TCP      | https > 53646 [ACK] Seq=2066 Ack=871 win=7424 Len=0                  |

|   |                                                                                                           |
|---|-----------------------------------------------------------------------------------------------------------|
| + | Frame 61 (1228 bytes on wire, 1228 bytes captured)                                                        |
| + | Ethernet II, Src: Cisco-Li_83:8f:7e (00:13:10:83:8f:7e), Dst: AsustekC_5e:ae:2f (00:22:15:5e:ae:2f)       |
| + | Internet Protocol, Src: 193.137.8.173 (193.137.8.173), Dst: 192.168.1.102 (192.168.1.102)                 |
| + | Transmission Control Protocol, Src Port: https (443), Dst Port: 53646 (53646), Seq: 1, Ack: 91, Len: 1174 |
| - | Secure Socket Layer                                                                                       |
| + | SSLv3 Record Layer: Handshake Protocol: Server Hello                                                      |
| + | SSLv3 Record Layer: Handshake Protocol: Certificate                                                       |
| + | SSLv3 Record Layer: Handshake Protocol: Server Key Exchange                                               |
| + | SSLv3 Record Layer: Handshake Protocol: Server Hello Done                                                 |

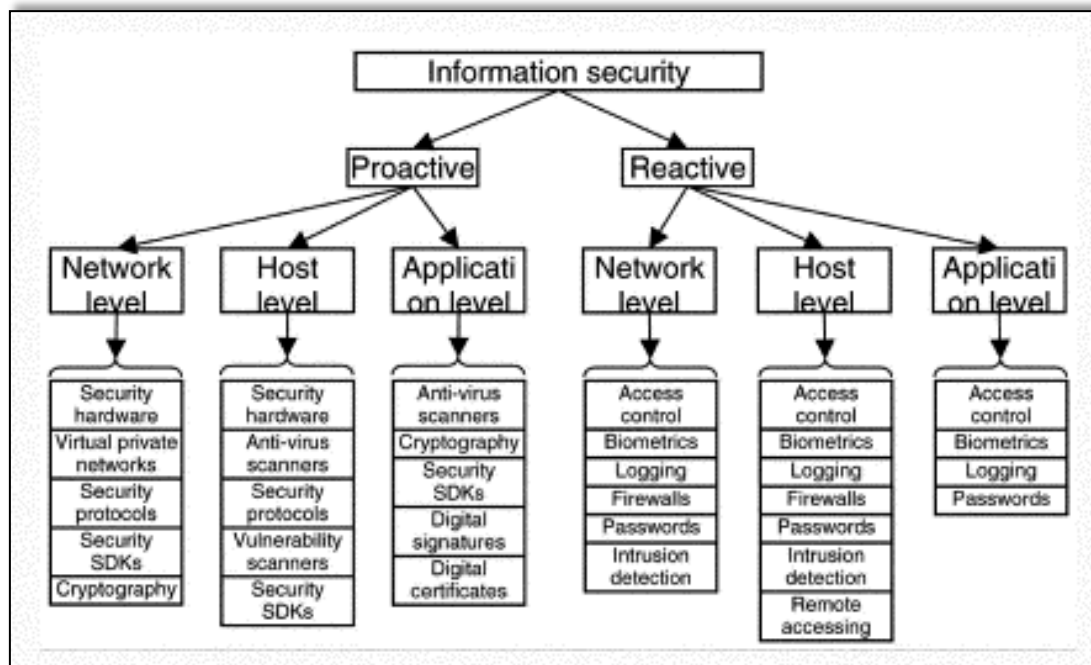
# Notes about TLS/SSL



# Notes about SSH

- SSH (Secure Shell) – Protocol that allows authentication and encrypted communication at the level of the operating system shell. Based on the client-server model.
- Originally for Unix, but already available for Windows
- Supports a (large) subset of the shell commands
- Replaces, with advantage, the traditional remote login utilities (**telnet**, **rlogin** or **rsh**)
- Cipher algorithm negotiation (DES, IDEA, AES,...)
- Authentication (includes support for PKIs and Kerberos)
- **Very exposed to Brute-forcing attacks**

# Taxonomy – InfoSec technologies



# Resources dedicated to each technology

| Resource                                | Access control | Biometrics | Remote access | Passwords | Cryptography | Digital signatures | Digital certificates | Firewalls | Virtual private networks | Intrusion detection systems | Vulnerability scanners | Anti-virus scanners | Security SDKs | Logging | Security protocols | Security hardware |
|-----------------------------------------|----------------|------------|---------------|-----------|--------------|--------------------|----------------------|-----------|--------------------------|-----------------------------|------------------------|---------------------|---------------|---------|--------------------|-------------------|
| <b>Journals</b>                         |                |            |               |           |              |                    |                      |           |                          |                             |                        |                     |               |         |                    |                   |
| Computers & Security [7]                | ✓              | ✓          | ✓             | ✓         | ✓            | ✓                  | ✓                    | ✓         |                          | ✓                           |                        | ✓                   | ✓             | ✓       | ✓                  |                   |
| Computer Fraud & Security [8]           |                | ✓          |               | ✓         | ✓            | ✓                  | ✓                    |           |                          | ✓                           | ✓                      | ✓                   |               | ✓       |                    | ✓                 |
| Network Security [9]                    |                | ✓          | ✓             |           | ✓            | ✓                  | ✓                    | ✓         | ✓                        | ✓                           | ✓                      | ✓                   | ✓             | ✓       | ✓                  | ✓                 |
| <b>Books</b>                            |                |            |               |           |              |                    |                      |           |                          |                             |                        |                     |               |         |                    |                   |
| Internet & TCP/IP Network Security [10] |                |            | ✓             |           |              |                    |                      | ✓         |                          |                             | ✓                      |                     |               |         | ✓                  |                   |
| Secure Communicating Systems [11]       |                |            |               |           | ✓            |                    |                      |           |                          |                             |                        |                     |               |         |                    |                   |
| Computer Security Policies [12]         |                |            |               |           |              | ✓                  | ✓                    | ✓         |                          |                             |                        |                     |               | ✓       |                    |                   |
| Windows 2000 Security [13]              | ✓              |            | ✓             |           | ✓            | ✓                  | ✓                    | ✓         |                          |                             |                        |                     | ✓             | ✓       | ✓                  | ✓                 |
| Hackers Beware [14]                     |                |            |               | ✓         |              |                    |                      |           |                          |                             | ✓                      |                     |               | ✓       |                    |                   |
| Computer Security [15]                  | ✓              |            |               | ✓         | ✓            |                    |                      |           |                          |                             |                        | ✓                   |               | ✓       |                    |                   |
| Hacking Exposed [16]                    |                |            | ✓             | ✓         |              |                    |                      | ✓         |                          | ✓                           | ✓                      | ✓                   | ✓             | ✓       |                    |                   |
| Intrusion Detection [17]                |                |            |               |           |              |                    |                      |           |                          | ✓                           | ✓                      |                     |               | ✓       |                    | ✓                 |
| Network Intrusion Detection [18]        |                |            |               |           |              |                    |                      |           |                          | ✓                           | ✓                      | ✓                   |               |         | ✓                  |                   |
| Access Denied [1]                       | ✓              | ✓          |               | ✓         | ✓            | ✓                  |                      |           | ✓                        |                             |                        | ✓                   |               |         |                    |                   |
| Internet & Intranet Security [19]       |                |            |               |           |              | ✓                  |                      | ✓         |                          |                             | ✓                      |                     |               |         | ✓                  |                   |
| Secrets & Lies [20]                     |                | ✓          |               |           | ✓            |                    |                      |           |                          |                             | ✓                      | ✓                   | ✓             |         |                    | ✓                 |
| Security Architecture [3]               | ✓              |            |               |           | ✓            |                    | ✓                    | ✓         | ✓                        | ✓                           | ✓                      |                     | ✓             | ✓       |                    | ✓                 |
| Security in Computing [21]              | ✓              |            |               |           | ✓            | ✓                  |                      | ✓         |                          |                             |                        | ✓                   |               |         |                    | ✓                 |
| Computer Security [22]                  | ✓              |            |               | ✓         |              | ✓                  |                      |           |                          |                             |                        | ✓                   | ✓             |         |                    |                   |
| Information Security Architecture [23]  | ✓              | ✓          | ✓             | ✓         |              |                    |                      | ✓         | ✓                        |                             |                        | ✓                   |               |         |                    |                   |
| Web Security [24]                       | ✓              |            | ✓             | ✓         |              | ✓                  | ✓                    | ✓         |                          |                             |                        |                     | ✓             | ✓       | ✓                  |                   |
| Web Security [25]                       |                |            | ✓             |           |              | ✓                  | ✓                    |           |                          |                             |                        | ✓                   | ✓             |         | ✓                  |                   |

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